

M.Tech Programmes

Curriculum (Highlights)

M.Tech in Computer Science and Engineering

M.Tech in Computer Science and Information Security

M.Tech in Computer Aided Structural Engineering

M.Tech in Product Design and Management (Applicable from 2024 batch)

M.Tech in VLSI & Semiconductor Technologies (Applicable from 2026 batch)

M.Tech in Computer Science and Engineering

The objective of the two-year M.Tech (CSE) programme is to produce post-graduates with advanced knowledge in one or more areas of Computer Science. The programme is designed such that a student can complete it based on advanced coursework alone. However, the students are given the facility to concentrate on a problem by substituting some of the courses with equivalent project work.

The first semester of the M.Tech programme is aimed at providing the necessary breadth in Computer Science and at integrating the students with the instructional philosophy of IIIT, Hyderabad. The subsequent semesters provide avenues for specializing in one or more of the streams. The semester wise distribution of courses is given below.

Semester	Course Name	Credits
1	Data Structures & Algorithms for Problem Solving	6
	Maths for Computer Science 1 – Probability & Statistics (H1)	2
	Maths for Computer Science 2 – Linear Algebra (H2)	2
	Software Systems Development	4
	Advanced Operating Systems	4
	Sem1 Total Credits	18
2	Bouquet Core	4
	Bouquet Core	4
	Area Elective	4
	Breadth Elective / Project / Independent Study	4
	Sem2 Total Credits	16
3	Bouquet Core	4
	Bouquet Core	4
	Area Elective	4
	Open Elective / Project / Independent Study	4
	Sem3 Total Credits	16
4	Area Elective	4
	Area Elective	4
	Area Elective	4
	Open Elective / Project / Independent Study	4
	Sem4 Total Credits	16
	Total Credits	66

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M. Tech in Computer Science and Information Security

The objective of the two-year M.Tech(CSIS) programme is aimed at producing the much needed highly skilled manpower in the area of Information Security, at the same time well versed with fundamentals of Computer Science.

Semester	Course Name	Credits
1	Data Structures & Algorithms for Problem Solving	6
	MCS1- Probability and Statistics (H1)	2
	MCS2- Linear Algebra (H2)	2
	Software Systems Development	4
	Advanced Operating Systems	4
	Sem1 Total Credits	18
2	System and Network Security	4
	Principles of Information Security	4
	Bouquet Elective	4
	Bouquet/ Area/ CS/ IT Elective	4
	Sem1 Total Credits	16
3	Research in Information Security	4
	Advanced Computer Networks	4
	Bouquet Elective	4
	Area/CS/IT Elective	4
	Sem1 Total Credits	16
4	Information Security Audit and Assurance	4
	PG Project	4
	Bouquet Elective	4
	Area/CS/IT Elective	4
	Sem1 Total Credits	16
	Total Credits	66

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M. Tech in Computer Aided Structural Engineering

Computer Aided Structural Engineering is an innovative program, focused on the combination of recent advances made in the field of structural engineering and computer science. This combination allows structural engineer the flexibility and freedom for a better understanding of structural behaviour with material and geometric nonlinearity and loading uncertainties. The students join this program with a background in civil engineering. The course provides an excellent grounding in the fundamentals of structural engineering subjects. It also engages in a comprehensive study of computer science subjects such as programming, databases, graphics, visualization etc. The objective of M.Tech CASE program is to include recent advances in the development and use of computer methods for the solution of scientific and engineering problems related to structures.

The objectives of the curriculum, while maintaining the rigour of the program are:

- to keep students abreast with the latest in Structural Engineering
- to improve mathematics skills of students;
- to introduce to students use of latest technology in structural engineering field;
- to familiarise students with the properties and behaviour of new and emerging materials used in construction; and
- to benefit domain as well as non-domain students through the program

Curriculum

Semester	Course Name	Credits
1	Structural Dynamics	4
	Computer Problem Solving	4
	FEM	4
	Structural Engineering Design Studio	4
	Sem1 Total Credits	16
2	Computing Tools	4
	Elective I	4
	Elective II	4
	Elective III	4
	Elective IV	4
	Sem2 Total Credits	20
3	IoT	4
	Elective V	4
	Elective VI	4
	Elective VII	4
	Elective VIII (half-course)	2
	Sem3 Total Credits	18

Semester	Course Name	Credits
4	Project (Full sem)	12
	Sem4 Total Credits	12
	Total Credits	66

List of Electives

Structural Engineering (SE)	Others (Civil Engineering)*
Earthquake Resistant Design	Spatial Informatics
Advanced Structural Analysis	Building Automation
Stability of Structures	Green Building
IS Codes on Design and Structural Safety Assessment	Disaster Management
Advanced Structural Design	Multivariate Analysis
Computational Earthquake Engineering	Remote Sensing
Structural Wind Engineering	Design of Hydraulic Structures
Probabilistic Methods in Structural Engineering	Illumination Engineering
Bridge Engineering	Hydro informatics
Earthquake Resistant Design of Masonry Structures	
Plastic Theory of Structures	
Seminar (half-course)**	

*The list of electives is only indicative.

** Seminar constitutes:

1. Student presentations on given subject area/research papers
2. Group discussions
3. Talks by industry experts
4. Writing technical papers/term papers

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M.Tech in Product Design and Management

(Applicable from 2024 batch)

About the Programme

M. Tech in Product Design and Management (PDM) is a program that converges technology trends in software, design thinking, user centric design, products & markets, business models around technology products. The program is being supported in-part by Ministry of Electronics and Information Technology, Government of India. It is a 2-year Masters programme.

An applicant to this programme is required to have a Bachelor's degree in Engineering/Technology/Design/Science or Equivalent (4 year programme after 10+2) or Masters degree in Science/Computer Applications (2 year programme after 10+2+3) with a minimum of 60% or equivalent grade. The course shall focus on ICT technology-oriented product design and management. Hence candidates who apply are expected to have at least 1 yr experience in product development (especially ICT related products). Some exposure to design is preferred (engineering design or hardware design or industrial design) but not necessary.

The program intake shall be 25-30 students. Scholarship amount shall be paid to some of the students to cover living expenses as per MeitY project rules. The candidate must be enrolled under full-time M.Tech PDM programme to be eligible for stipend. The two-year program contains a mix of Technology, Entrepreneurship and Design courses.

Key Objectives

The aim of M. Tech in Product Design and Management program is to create market and design aware technologists that are equipped with knowledge needed to be able to conceptualize, define, design technology products and connect them with markets. The curriculum shall create graduates that understand the convergence of technology, design, markets and the business of products. The program shall enable creation of new products, new startups, groom early career IT professions to become technology product managers for technology companies specifically around information technologies.

The semester wise distribution of courses is given below.

Courses	Code	Credits
Core Courses		22
Design Thinking - Research to Define	PD1.301	2
Design Thinking - Idea to Evaluate	PD1.401	2
User Research Methods	CS9.501	2
Product Management 101	PD2.401	2
Business Fundamentals	PD2.421	4
Business Finance	PD2.422	4
Lab to Land: Product viability (title to be given)	TBD	2
Software Engineering (SE)	CS6.401	4
Design Electives		6 - 10
Human Computer Interaction (HCI)	PD1.501	2
Product Design Workshop	PD1.411	4
User Interaction and Usability of Digital Products	CS5.402	4
Business Electives		6 - 10
Technology Product Entrepreneurship	CS9.424	4
Organizational Operations	PD2.423	2
Product Marketing	PD2.501	4
Product Lifecycle Management	PD2.502	4
System Electives**		8 -12
Bridge Courses***		0-4
Software Systems Development (SSD)	CS6.302	4
Open Electives****		0 - 8
PDM Final Project		16
Product Development Project – (Part 1) *****	PD9.403	8
Product Development Project – (Part 2) *****	PD9.404	8
Industry/Institute Seminar*****		0

** A list of System elective courses shall be made available based on the semester they are offered. Only course code above 400 can be taken with the consent of the Program Head.

*** Students must take the bridge course if they do not pass the exemption exam prior to the start of the semester.

**** Any course offered at IIITH (course code above 400) may be taken as an Open Elective with the consent of the program head

***** Students must complete minimum of 28 credits with an overall CGPA of 7.0 or above before they can start the product development project. Any exemption to this requires the approval of the Program Head and/or Dean Academics. Only those students who has received a minimum of “C” grade in PDM Final Project Part 1 are eligible to register for PDM Final Project Part 2.

***** Students must “Pass” the “o” credit Industry/Institute seminar course for two semesters. It is recommended that this be taken in the first and second semester.

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M.Tech in VLSI & Semiconductor Technologies

(Applicable from 2026 batch)

About the Programme

M.Tech in **VLSI and Semiconductor Technologies** is a **2-year program** that integrates advanced VLSI design, semiconductor device fabrication, and industry-relevant research. The program leverages state-of-the-art laboratories at IIIT Hyderabad and University of Hyderabad (UoH), along with collaborations with industry, to produce engineers capable of solving real-world semiconductor challenges.

Applicants should hold a **Bachelor's degree in Electronics, Electrical, Computer, or related Engineering/Technology** or a **Master's degree in Science/Computer Applications** with a minimum of 60% or equivalent grade. GATE qualification in ECE/EE/IN is preferred but not mandatory. Candidates with prior exposure to IC design, embedded systems, or semiconductor devices are preferred but not mandatory.

The program intake shall be **30 students per year**, and selected students may receive financial support through scholarships or project-based stipends. The curriculum combines **core semiconductor technology, advanced VLSI design, cleanroom device fabrication, EDA-based design, and research-driven projects**.

Key Objectives

The program aims to:

1. Deliver a research-intensive curriculum that builds deep expertise in VLSI design, semiconductor devices, and fabrication technologies.
2. Integrate industry-relevant workflows, including full-cycle IC design, verification, layout, tape-out, and testing using state-of-the-art EDA tools and PDKs.
3. Provide hands-on training in cleanroom fabrication and device characterization, aligned with national semiconductor priorities.
4. Enable innovation through research projects, industry-driven problem statements, internships, and collaborations with semiconductor companies and R&D labs.
5. Foster interdisciplinary skills across MEMS, RFICs, device modelling, packaging, emerging semiconductor and AI hardware technologies.
6. Combine academic rigor with industry mentoring to produce graduates who are both research-capable and industry-ready.
7. Prepare students for advanced R&D careers, Ph.D. pathways, and technology leadership roles through a strong thesis and publication-oriented research component.

Curriculum

Semester I

Course Name	Credit
Area Elective*	4
Area Elective	4
Area Elective	4
Area Elective	4

Semester 2

Course Name	Credit
Area Elective	4
Area Elective	4
Area Elective/Independent study	4
Area Elective/Independent study	4

Course Name	Credit
Summer Project	2

Semester 3

Course Name	Credit
Open Elective	4
Area Elective	4
M.Tech Project	4
M.Tech Project	4

Semester 4

Course Name	Credit
M.Tech Project	4
M.Tech Project	4
M.Tech Project	4
M.Tech Project	4

* Area Electives:

Area Electives are broadly classified under following three categories covering various domains such as Digital, Analog, RF, Mixed-Signal, EDA, Computer-Architecture, Nanotechnology, Device physics, Device fabrication and Flexible Electronics.

A student must complete 1 course from each of these 3 categories, and at least 3 courses in one of these three categories.

Category-I: VLSI Design: Analog IC Design, Digital VLSI Design, Principles of Semiconductor Devices, CMOS References and Regulators, CMOS Oscillators

Category-II: Embedded Systems & AI: Digital VLSI Design, Advanced Comp Architectures, Compilers, Hardware for AI, Software for performance

Category-III: Semiconductor Technologies: Principles of Semiconductor Devices, Advanced Semiconductor Devices, Flexible electronics, Instrumentation & Devices, Nano Fabrication Lab, Microelectronics Simulation Lab, III-V Compound Semiconductors, Nano Technology, MEMS and THz Technology, Optoelectronics

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